

EECE 571Q

Lecture 01: What could possibly go wrong?

Tuesday 12 May 2026

Quantum computers today

Termed **noisy, intermediate scale quantum** (NISQ) devices.

Quantum Computing in the NISQ era and beyond

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Eprint: [arXiv:1801.00862v3](https://arxiv.org/abs/1801.00862v3)

Doi: <https://doi.org/10.22331/q-2018-08-06-79>

Citation: Quantum 2, 79 (2018).

Quality is steadily improving in the face of many challenges

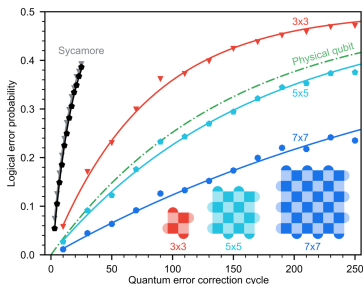
- Short coherence times
- High error rates (single-qubit, two-qubit operations)
- Limited qubit connectivity
- Scaling up the hardware

Quantum computing beyond the NISQ era

Solving problems *at scale* entails **quantum error correction** and **fault-tolerant quantum computing**.

People are actually doing it!

Example: Google's *Willow* (105 superconducting qubits).



How does it work, and *why* does it work?

Image: <https://research.google/blog/making-quantum-error-correction-work/>

Course learning outcomes

Core goal: gain a holistic understanding of what's required to implement a fault-tolerant quantum algorithm from end to end.

- describe the role and importance of error correction and fault-tolerance in quantum computing, and the current state of progress in the field
- apply encoding and decoding strategies, and logical operations, to execute quantum algorithms using one or more error-correcting codes, both mathematically and in software
- perform simulations to estimate the threshold of quantum error-correcting codes
- demonstrate mastery of a chosen topic in QECC / FTQC by developing and delivering lectures to your peers

Contents

Divided into 2 modules:

- ➊ Quantum error correction (lectures 1-6)
- ➋ Fault-tolerant quantum computing (lectures 7-12)

Suggested textbooks:

- [Surviving as a Quantum Computer in a Classical World](#), Daniel Gottesman
- Quantum Computation and Information, Nielsen and Chuang (available from bookstore/library)
- [Understanding Quantum Information and Computation](#), John Watrous (also on [YouTube](#))
- [PennyLane Codebook](#)

Logistics

My info:

Office:	KAIS 3043
Office hrs:	T 11:00-12:00 or by appointment. Open-door policy after 12:00 weekdays
Email:	olivia.dimatteo@ubc.ca
GitHub:	glassnotes

All lecture slides/demos posted on course GitHub:

<https://github.com/glassnotes/EECE-571Q-FTQC>

Assignments and grading

Three components:

- 10% in-class quizzes
- 30% mid-term exam (during S1 exam period)
- 60% final project

Assignments and grading, cntd.

10% quizzes (5-10):

- First one Tuesday; in class, in PrairieLearn
- Individual, but open book/notes/docs; **AI use prohibited**
- Math and conceptual questions based on previous week's material

Purpose is for reviewing content. Schedule is in syllabus.

Your lowest quiz grade will be automatically dropped.

Quiz questions will be made available for practice after class.

Assignments and grading, cntd.

30% midterm:

- during T1 exam period (22-26 June)
- pen-and-paper questions
- closed-book and closed-notes (allowed one pre-approved, hand-written cheat sheet - details provided closer to date)

No assignments, but supplementary exercises will be provided to explore beyond in-class material and help with midterm preparation.

Assignments and grading, cntd.

60% final project: deep dive on a specialized topic of your choice

- Work individually or as a pair
- Deliver 2 lectures
- Evaluation:
 - (10%) Proposal and lecture outline
 - (65%) Lecture (typeset notes, slides, in-class delivery)
 - (10%) In-class quiz design
 - (10%) Software implementation / demonstrations
 - (5%) Reflection

Example project topics in syllabus; more details in coming weeks.

Academic misconduct and AI usage policies

The following will lead to a grade of 0 and/or a misconduct investigation:

- copying the work of classmates
- written content that closely paraphrases external sources
- unauthorized use of generative AI tools for quizzes

Exceptions:

- if your IDE has, e.g., Copilot integration, must acknowledge as source in assignment submission
- can use for light editing of *pre-written text* for lecture notes; must submit usage log

What do all these numbers mean?

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You do not have access to this system in this account. Switch accounts or [explore plans](#)

Details

Qubits	Couplers	2Q error (median)	2Q error (layered)	CLOPS
156	176	1.17E-3	2.09E-3	340K
Status	Region	QPU version	Processor type	
Online	Washington DC (us-east)	1.0.4	Heron r3	
Basis gates	Pending jobs	2Q error (best)	Readout error (median)	
c _z , id, rx, rz, rz _z , sx, x	174	6.33E-4	5.43E-3	
CZ error (median)	SX error (median)	T1 (median)	T2 (median)	
1.172E-3	1.443E-4	290.2 us	330.77 us	

IBM Quantum Platform (Screenshot 2026-04-30)

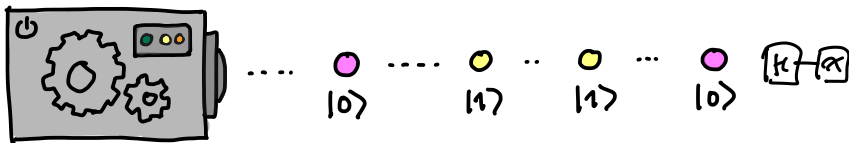
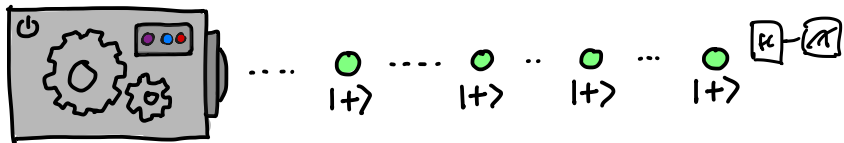
Learning outcomes

Part 1: quantum error correction

Today's lecture: *"What could possibly go wrong?"*

- Perform quantum computation in the density matrix formalism
- Evaluate the outcome of quantum error channels on input states
- Describe, and experimentally estimate, processor quality metrics

Review: pure vs. mixed states



Review: density matrices

Probabilistic mixtures of states (**mixed states**) are represented by **density matrices**:

$$\{p_i, |\psi_i\rangle\} \longrightarrow \rho = \sum_i p_i |\psi_i\rangle\langle\psi_i|$$

$$\sum_i p_i = 1$$

Exercise: A system prepares $|+\rangle$ with probability $1/3$, and $|0\rangle$ with probability $2/3$. What is its state?

$$\begin{aligned} \rho &= \left(\frac{\frac{1}{3} + \frac{2}{3}}{6} \quad \frac{1}{6} \right) = \frac{1}{6} \begin{pmatrix} 5 & 1 \\ 1 & 1 \end{pmatrix} \\ &= \frac{2}{3} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + \frac{1}{3} \cdot \frac{1}{2} \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \\ &= \frac{2}{3} |0\rangle\langle 0| + \frac{1}{3} |+\rangle\langle +| \end{aligned}$$

Review: density matrices

Density matrices have some nice properties.

- they are Hermitian
- they have trace 1
- they are positive semi-definite (all eigenvalues are ≥ 0)
- (for pure states only) they are projectors, i.e., $\rho^2 = \rho$

$$\rho = |\psi\rangle\langle\psi| \rightarrow \rho^2 = |\psi\rangle\langle\psi| \psi\rangle\langle\psi| = |\psi\rangle\langle\psi|$$

Check with our example:

$$\rho = \begin{pmatrix} 5/6 & 1/6 \\ 1/6 & 1/6 \end{pmatrix}$$

$\sum = 1$

$$\lambda_1 = 0.873 \geq 0$$

$$\lambda_2 = 0.127 \geq 0$$

Review: unitary operations and mixed states

Exercise: Apply H to a qubit in the mixed state $|+\rangle$ w/prob. $1/3$, $|0\rangle$ w/prob. $2/3$. What state is the qubit in now?

$$\begin{aligned}
 H \rho H^\dagger &= H \left[\frac{1}{3} |+\rangle\langle+| + \frac{2}{3} |0\rangle\langle 0| \right] H \\
 &= \frac{1}{3} H |+\rangle\langle+| H + \frac{2}{3} H |0\rangle\langle 0| H \\
 &= \frac{1}{3} |0\rangle\langle 0| + \frac{2}{3} |+\rangle\langle+| \\
 &= \begin{pmatrix} \frac{1}{3} & 0 \\ 0 & 0 \end{pmatrix} + \begin{pmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & \frac{1}{3} \end{pmatrix} \\
 &= \frac{1}{3} \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}
 \end{aligned}$$

Review: unitary operations and mixed states

For pure state $|\psi\rangle$ and operation U ,

$$|\psi\rangle \rightarrow |\psi'\rangle = U|\psi\rangle$$

As mixed states,

$$|\psi\rangle\langle\psi| \rightarrow |\psi'\rangle\langle\psi'| = U|\psi\rangle\langle\psi|U^\dagger$$

More generally,

$$\begin{aligned} \rho' &= U \rho U^\dagger \\ &= U \left(\sum_i p_i |\psi_i\rangle\langle\psi_i| \right) U^\dagger \\ &= \sum_i p_i U |\psi_i\rangle\langle\psi_i| U^\dagger \end{aligned}$$

Review: mixed states and measurements

If we prepare a system in pure state $|\psi\rangle$ and measure in the basis $\{|\varphi_k\rangle\}$, the probability of observing it in state $|\varphi_k\rangle$ is

$$\begin{aligned}\text{Pr}(\text{outcome } k) &= |\langle\varphi_k|\psi\rangle|^2 \\ &= \langle\varphi_k|\psi\rangle \cdot \langle\psi|\varphi_k\rangle \\ &= \langle\psi| \boxed{|\varphi_k\rangle\langle\varphi_k|} |\psi\rangle \\ &= \langle\psi| P_{\varphi_k} |\psi\rangle\end{aligned}$$

$M_k = |\varphi_k\rangle\langle\varphi_k|$ *projects* state $|\psi\rangle$ onto the state $|\varphi_k\rangle$.

Review: mixed states and measurements

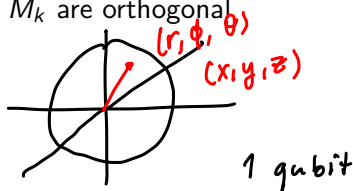
Let $M = \{M_k\}$ be positive semi-definite Hermitian operators in a Hilbert space. M_k is associated to measurement outcome k .

M is a **positive operator-valued measure (POVM)** if

$$\sum_k M_k = I$$

M is a **projective measurement** if all M_k are orthogonal projectors, i.e.,

- $M_k^2 = M_k$
- $M_k M_j = \delta_{kj} M_k$



Review: mixed states and measurements

The probability of outcome k is computed using the **Born rule**:¹

$$\text{Pr}(k) = \text{Tr}(M_k \rho)$$

The expectation value of a Hermitian operator, B , is

$$\langle B \rangle = \text{Tr}(B \rho)$$

Practice exercise: Derive $\langle B \rangle$ starting from the fact that an expectation value is a weighted average over the (eigen)values assigned to measurement outcomes of B , i.e.,

$$\langle B \rangle = \sum_k \lambda_k \text{Prob}(k)$$

¹You can show this is equivalent to a projective measurement when ρ is pure

Review: mixed states and measurements

Exercise: Suppose we prepare our system in $|+\rangle$ with probability $1/3$ and $|0\rangle$ with probability $2/3$. What is the probability of obtaining the POVM outcome $\Pi_+ = |+\rangle\langle+|$?

$$\begin{aligned} \Pr(+)&= \underbrace{\text{Tr}\left(|+\rangle\langle+| \cdot \frac{1}{3}|+\rangle\langle+|\right)}_{\frac{1}{3}} + \underbrace{\text{Tr}\left(\frac{2}{3}|+\rangle\langle+|0\rangle\langle 0|\right)}_{\frac{1}{3}} \\ &= \frac{2}{3} \end{aligned}$$

Review: mixed states and measurements

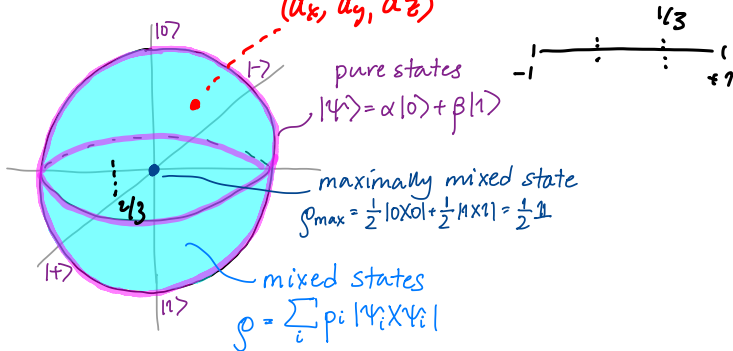
Exercise: Suppose we prepare our system in $|+\rangle$ with probability $1/3$ and $|0\rangle$ with probability $2/3$. What is $\langle X \rangle$?

$$\begin{aligned}\langle X \rangle &= \text{Tr}(X\rho) \\ &= \text{Tr}\left(\frac{1}{3}\underbrace{X|+\rangle\langle+|} + \frac{2}{3}\underbrace{X|0\rangle\langle 0|}\right) \\ &= \text{Tr}\left(\frac{1}{3}|+\rangle\langle+| + \frac{2}{3}\underbrace{|1\rangle\langle 0|}\right) \\ &= \frac{1}{3}\end{aligned}$$

no diag.

Expectation values of Paulis relate to Bloch sphere coordinates.

$$(a_x, a_y, a_z)$$



Practice exercise: derive relationship and Bloch vector properties (see PrairieLearn).

Summary

We can now

- represent quantum states
- apply unitary operations
- perform measurements

... how do we model errors?

Quantum channels

Operations, measurements, and errors are **quantum channels**.

We are most often interested in channels that map quantum states (density matrices) to other density matrices.

$$\Phi: \mathcal{H}_1 \rightarrow \mathcal{H}_2 \quad \rho' = \Phi(\rho)$$

Quantum channels are completely positive, trace-preserving (CPTP) linear maps.

- Trace-preserving: $\text{Tr}(\rho') = \text{Tr}(\rho)$
- Positive: $\rho' \geq 0$
- Completely positive: $(\Phi \otimes I_n)$ must be positive for all n

Quantum channels

Quantum channels are characterized by **Kraus operators** $\{K_i\}$,

$$\Phi(\rho) = \sum_i K_i \rho K_i^\dagger \quad \sum_i K_i^\dagger K_i = I$$

Kraus operators represent, loosely, a “set of things the channel can do to a system”.

Example: a channel with one Kraus operator is a unitary channel.

$$U_i^\dagger U_i = I \quad \mathcal{U}(\rho) = U_i \rho U_i^\dagger$$

Measurement as a quantum channel

Measurement effects the following transformation:

$$\rho' = \sum_k \underbrace{\text{Prob}(k)} \cdot \underbrace{\text{State after observing outcome } k}$$

Given measurement operators $\{M_j\}$, write $M_j = \underbrace{K_j^\dagger K_j}$. Then,

$$\sum_j M_j = I \rightarrow \sum_j K_j^\dagger K_j = I$$

Hermitian

$$(K_j^\dagger K_j)^\dagger = K_j^\dagger K_j$$

Full channel effect is

$$\Phi(\rho) = \sum_j K_j \rho K_j^\dagger = \sum_j \text{Prob}(j) \cdot \underbrace{\frac{K_j \rho K_j^\dagger}{\text{Prob}(j)}}_{\text{state after meas.}}$$

Measurement as a channel

Concrete example: $\{M_j\} = \{\Pi_+, \Pi_-\}$:

Consider mixed state

Intuitively, after measurement expect

Error channels

Errors are modelled using quantum channels,

$$\Phi_E(\rho) = \sum_i E_i \rho E_i^\dagger$$

Think of Kraus operators as possible errors $\{E_i\}$ that occur with probability $\text{Tr}(E_i \rho E_i^\dagger)$.

Example: A bit flip (Pauli X) error occurs with probability p .

$$\Phi_{\text{BF}}(\rho) = (1-p)\rho + p X \rho X^\dagger$$

$$K_0 = \sqrt{1-p} I \quad K_1 = \sqrt{p} X$$

Bit flip channel

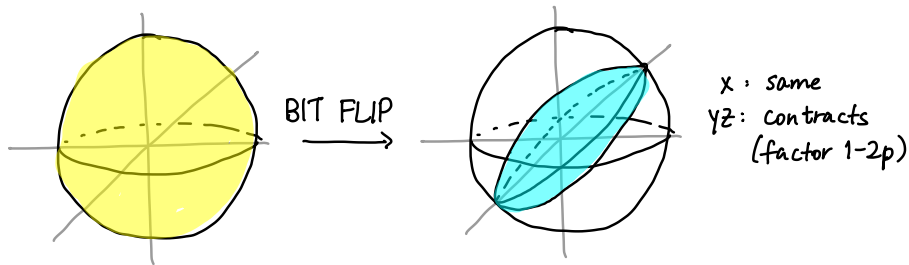
Exercise: explicitly evaluate $\Phi_{BF}(\rho)$ for

↑
Take home

$$\rho = \begin{pmatrix} a & b \\ b^* & 1-a \end{pmatrix}$$

and the expectation values $\langle X \rangle$, $\langle Y \rangle$, and $\langle Z \rangle$. What is the effect of the channel on the state's Bloch vector?

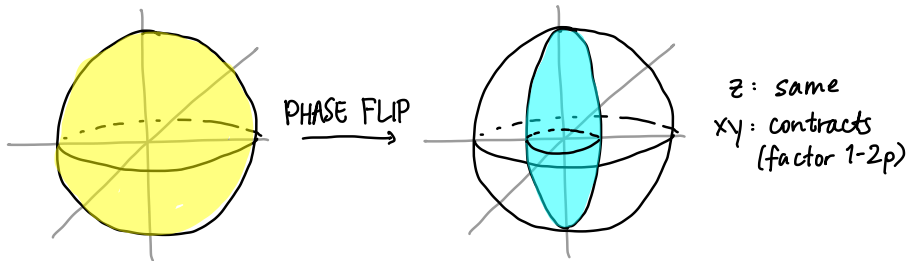
Bit flip channel



Dephasing channel

$$\Phi_{PF}(\rho) = (1-p)\rho + p Z\rho Z$$

$$K_0 = \sqrt{1-p} I \quad K_1 = \sqrt{p} Z$$



Practice exercise: work with an alternative specification of Kraus operators for the dephasing channel (see PrairieLearn).

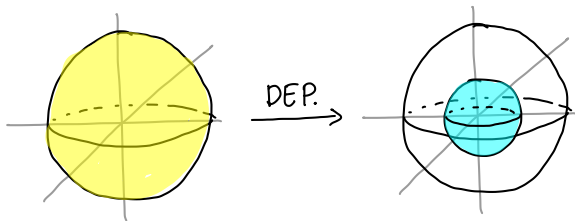
The depolarizing channel

Each Pauli error occurs with equal probability,

$$\Phi_D(\rho) = (1-p)\rho + \frac{p}{3}X\rho X + \frac{p}{3}Y\rho Y + \frac{p}{3}Z\rho Z$$

Φ_D can also be written as

$$\Phi_D(\rho) = (1-\lambda)\rho + \lambda \frac{I}{2}$$



How bad is it?

★ We will start here next time.

One approach for quantifying the strength and effects of noise is to define mathematical distance functions between density matrices.

A common metric is the **trace distance**:

Bounded by $0 \leq T(\rho, \sigma) \leq 1$; *lower* trace distance is better.

How bad is it?

Can also compare using **fidelity**,

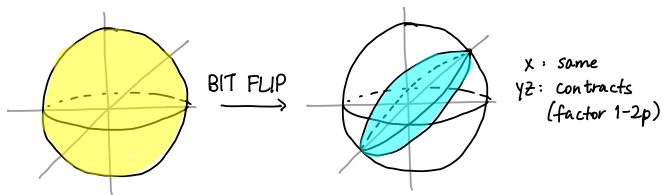
Bounded by $0 \leq F(\rho, \sigma) \leq 1$. *Higher* fidelity is better.

Special case: if ρ is pure but σ is mixed,

Practice exercise: derive this and other special cases (see PrairieLearn).

How bad can it get?

Exercise: Determine $F(\rho, \sigma)$ if ρ is pure and $\sigma = \Phi_{BF}(\rho)$ with parameter p .



How bad can it get?

Exercise: What happens if the strength of depolarizing noise is $p = 3/4$? What is the fidelity of any pure ρ with the output of such a channel?

Amplitude damping channel

Suppose $|1\rangle$ relaxes to $|0\rangle$ with probability p .

How do you think this affects the Bloch sphere?

Amplitude damping channel

Amplitude damping is related to a different quality metric.

Assume probability of remaining in $|1\rangle$ decays with time,

Let's do some experiments to simulate qubit characterization.

Recap

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You do not have access to this system in this account. Switch accounts or [explore plans](#)

Details

Qubits 156	Couplers 176	2Q error (median) 1.17E-3	2Q error (layered) 2.09E-3	CLOPS 340K
Status ● Online	Region Washington DC (us-east)	QPU version 1.0.4	Processor type Heron r3	
Basis gates cz, id, rx, rz, rzz, sx, x	Pending jobs 174	2Q error (best) 6.33E-4	Readout error (median) 5.43E-3	
CZ error (median) 1.172E-3	SX error (median) 1.443E-4	T1 (median) 290.2 us	T2 (median) 330.77 us	

IBM Quantum Platform (Screenshot 2026-04-30)

Next time

Content:

- Correcting bit- and phase-flip errors
- Correcting arbitrary single-qubit errors
- Conditions for quantum error correction

Action items:

- ① Sign up for PrairieLearn and Piazza
- ② Start thinking about project topics
- ③ Supplementary exercises

Recommended reading:

- From today: Gottesman 1.1-1.3; Watrous 2.2-2.3; Codebook N, DM
- For next class: Gottesman 2, Codebook E